YASSER EL-SAYED M. EL-ALFY

Researcher | KFUPM

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Yasser El-Alfy is a researcher at KFUPM working on learning-based perception and 3D SLAM. His work focuses on integrating deep learning with traditional robotics to build robust navigation systems for autonomous vehicles and drones.

CITATIONS: 34 | H-INDEX: 3 | i10-INDEX: 1

EDUCATION

Doctor of Philosophy in Computer Engineering
King Fahd University of Petroleum and Minerals, Saudi Arabia
2023-present

Master of Science in Computer Engineering
King Fahd University of Petroleum and Minerals, Saudi Arabia
2021-2022

CGPA: 3.688

Bachelor of Science in Computer & Communication Engineering (BSCCE)
Ahlia University, College of Engineering, Bahrain
2018-2020

CGPA: 3.99

High School
Ahlia Schools, King Fahd University of Petroleum and Minerals, Saudi Arabia
2012-2014

Cumulative Grade: 99.92%

WORKSHOPS

Advanced Quantum Computing Workshop
Saudi Data and Artificial Intelligence Authority (SDAIA)
11-15 Jan 2026

PROFESSIONAL EXPERIENCE

Teaching and Lab Assistant
King Fahd University of Petroleum and Minerals (KFUPM)
2023-present

Research Intern
Center for Communication and Information Technology Research (CCITR), Research Institute, KFUPM
2020

Teaching and Lab Assistant | **Ahlia University**
2018-2020

RESEARCH STRATEGY

My research focuses on bridging the gap between raw sensor data and high-level spatial understanding. I develop monocular depth estimation models and integrate them into SLAM pipelines to enable 3D mapping without expensive LiDAR hardware.

Key Focus Areas:

- Smart Cities
- Autonomous Navigation
- Drone Applications
- Monocular Depth Estimation
- Visual SLAM & Navigation
- Anomaly Detection
- Road Anomaly Characterization
- Edge AI for Robotics
- 3D Mapping
- Scene Understanding

TECHNICAL SKILLS

Languages:

Java, JavaScript, Visual Basic, Python, TensorFlow, Pytorch, HTML, Assembly, Arduino, VHDL, Verilog

Software:

Matlab & Simulink, GNU Octave, Sumo Simulator, Cooja/Contiki Simulator, Xilinx, Modelsim, Cisco Packet Tracer, Arduino IDE, mikroc pro, ISIS 7 Professional, Eclipse, Visual Studio, Visual Studio Code, Google Colab, Jupyter Notebook, Microsoft office programs (Word, Excel, PowerPoint, Access), Latex professional editing, Overleaf, Adobe Photoshop, GIMP, Adobe Indesign, DaVinci Resolve, PowerDirector, Kdenlive, Windows Movie Maker, Microsoft Teams, Zoom, Google Meet, Cisco Webex, Blackboard, Moodle

OS:

Background in different Operating Systems: Windows, Mac, Linux

Other Skills:

Teamwork and communications skills, Research skills, Video and graphical editing

TEACHING ACADEMIC PROFILE

COE 412: Senior Design Project II — 2024-2026

COE 301: Computer Organization — 2024-2025

COE 292: Introduction to Artificial Intelligence — 2022-2024

ITCS 122: Introduction to Programming Techniques — 2018-2020

ITCS 101: Introduction to Computers & IT 2018-2020

ITCS 224: Data Structures 2019-2020

ITCS 201: Object Oriented Programming I 2019-2020

ITCS 121: Computer Programming 2018-2019

ITCS 222: Visual Programming 2018-2019

ITMS 325: Web Applications Design 2018-2019

TALKS & EVENTS

Multi-Modal Curriculum Federated Learning for Breast Cancer detection Seminar | 26th November 2025, 4:00PM

Bldg. 22, Room 119, King Fahd University of Petroleum and Minerals (KFUPM)

Combating Covid-19 Pandemic by Medical Face Mask Wearing Detection: Transfer Learning vs. Custom Architectures Seminar | 17th November 2021, 3:00PM

Bldg. 22, Room 134, King Fahd University of Petroleum and Minerals (KFUPM)

CERTIFICATIONS

IELTS Certificate: 6.5 2020

ICDL Certificates in Microsoft Office (Word, Spreadsheets, PowerPoint) 2018

Certificates of appreciation for being lab assistant, Ahlia Univ. 2018

Qiyas - General Aptitude Test: 100% 2014

Qiyas - Achievement Aptitude Test: 97% 2014

SELECTED PUBLICATIONS

Pothole detection and characterization: integrated segmentation and depth estimation in road anomaly systems
Neural Computing and Applications 38 (13), 540, 2026, (2026)

Enhancing pothole detection and characterization: integrated segmentation and depth estimation in road anomaly systems
Systems and Soft Computing (2025), (2025)

Monocular based 3D depth estimation and SLAM integration
Drone Systems and Applications (2025), (2025)

Intelligent Road Anomaly Detection with Real-time Notification System for Enhanced Road Safety
Sustainability (2025), (2025)

Real-Time On-Device Vision-Based Road Anomaly Detection with Frame Span Interval (FSI) Event Counting
IEEE AICCSA (2025), (2025)

Dynamic Real-time Learning of Electromagnetic Interference Parameters for Drone Crashing Attack
IEEE PIMRC (2025), (2025)

A Comparative Study of Transfer Learning and Custom Models with SSD for Face Mask Detection
IEEE (2025), (2025)

Pothole Detection Under Various Weather Conditions Using YOLOv9
Remote Sensing (2025), (2025)

Enhancing monocular depth estimation with an advanced encoder-decoder architecture
Neural Computing and Applications 37 (29), 24265-24280, (2025)

A federated learning approach to banking loan decisions
ISNCC (2023), (2023)

Distributed Intrusion Detection Systems Based on Deep Learning Techniques and Boosting Ensemble
ACM (2023), (2023)

Enhancing Monocular Depth Estimation Using an Enhanced Encoder--Decoder Model
Authorea Preprints, (2023)

Arduino based obstacle awareness vest for blind people
ICT-ASET (2022), (2022)

MONOCULAR DEPTH ESTIMATION USING DEEP LEARNING FOR 3D SLAM MAPPING IN AUTONOMOUS NAVIGATION
KFUPM Master Thesis, (2022)

Ultrasonic Projection for the Blind & Visually Impaired: A Proposed Model (Blind-Bat)
Ahlia University BSCCE Graduation Project, (2020)

PATENTS

SYSTEM AND METHOD FOR DETECTING AND COUNTING ROAD ANOMALIES US12445815

DATASETS

Road Anomaly Dataset with Depth Ground Truth
The first pothole dataset with depth ground truth values, manually annotated for high accuracy.

Pothole Detection Under Various Weather Conditions Dataset
A dataset with pothole images and generated/simulated various weather conditions.